Shahzad Ali

Date of birth: 12/08/1992 | **Nationality:** Pakistani | **Phone:** (+39) 3482657723 (Mobile) | **Email:** shahzadali039@gmail.com | **Email:** shahzad.ali6@unibo.it | **LinkedIn:** www.linkedin.com/in/shahzadali21 | **Scopus AuthorID:** 57202066536 | **WoS ResearcherID:** ODJ-8978-2025 | **WeChat:** shahzad-aliii | **ORCID:** https://orcid.org/0000-0002-0608-9515 |

Address: Via San Quirico, 25/15, 16163, Genova, Italy (Work)

ABOUT MYSELF

I'm a passionate PhD scholar specializing in the applications of Machine/Deep Learning in the realm of medical and healthcare. My work involves leveraging these technologies to address critical medical challenges, and I have a strong background in handling extensive datasets, developing algorithms, and building predictive models. I've also presented my research in reputed international journals and conferences. Additionally, I enjoy collaborating with peers and mentoring students.

EDUCATION AND TRAINING

01/11/2022 – CURRENT Bologna, Italy

PHD DATA SCIENCE AND COMPUTATION Alma Mater Studiorum Università di Bologna

Website https://www.unibo.it/it | **Field of study** Machine Learning Methods in Digital Health

01/2016 – 07/2018 Lahore, Pakistan

MASTER OF SCIENCE IN COMPUTER SCIENCE University of Engineering and Technology, Pakistan

Website https://uet.edu.pk/ | **Field of study** Computer Science | Machine Learning | **Final grade** 3.75 / 4.00 |

Thesis Artificial Neural Network Model for Rainfall-Runoff Prediction for Jhelum River Basin Upstream of the Mangla Dam, Pakistan

08/2011 – 06/2015 Sahiwal, Pakistan

BACHELOR OF SCIENCE IN TELECOMMUNICATION AND NETWORKING COMSATS University Islamabad, Sahiwal Campus

Website https://sahiwal.comsats.edu.pk/ | **Field of study** Computer Science | Telecommunication and Networking |

Thesis Visible Light-Based Communication System for High Data Rate Real-Time Applications

WORK EXPERIENCE

IRCCS OSPEDALE POLICLINICO SAN MARTINO – GENOVA, ITALY

Link https://www.ospedalesanmartino.it/it/

PHD RESEARCHER – 11/09/2023 – CURRENT

- Applied Machine Learning, Deep Learning, and Graph Networks for disease diagnosis using Neuroimaging data specifically focusing on multimodal data
- Developed AI models to enhance diagnostic accuracy for neurological disorders.
- Analyzed large-scale neuroimaging datasets to identify biomarkers and disease patterns.
- Collaborated with medical professionals and researchers for data-driven insights.
- Published research in top-tier journals and presented at international conferences.

DEPARTMENT OF INFORMATION SCIENCES, UNIVERSITY OF EDUCATION – LAHORE, PAKISTAN

Website https://ue.edu.pk/

LECTURER - COMPUTER SCIENCE – 03/2019 – CURRENT

Currently, teaching the following subjects at undergraduate level:

- Data Mining

- Artificial Intelligence
- Introduction to Machine Learning

Supervising final-year projects at undergraduate-level
Attending and speaking at conferences and seminars

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING, UNIVERSITY OF ENGINEERING AND TECHNOLOGY, LAHORE – LAHORE, PAKISTAN

Website https://uet.edu.pk/

GRADUATE RESEARCH ASSISTANT – 10/2016 – 05/2018

Taught and designed laboratory subject:

- Artificial Intelligence (Fall 2016, Fall 2017);

Assisted faculty members in creating and marking quizzes/assignments of undergraduate students and supervising final year and class projects.

PUBLICATIONS

2025

Graph Neural Networks in Alzheimer's Disease Diagnosis: A Review of Unimodal and Multimodal Advances

Authors: Shahzad Ali, Michele Piana, Matteo Pardini, and Sara Garbarino | **Journal Name**: Frontiers in Neuroscience (Impact Factor: 3.2)

2025

Evaluation of Post-Endodontic Pain Reduction Using Intracanal Cryotherapy in Symptomatic Apical Periodontitis

Authors: Anam Fayyaz Bashir, Ussamah Waheed Jatala, Muhammad Amber Fareed, Sheryar Sheryar, Saadia Ahmad Chattha, Saima Razaq Khan, Shahzad Ahmad, Shazia Iqbal, Muhammad Sohail Zafar, and Shahzad Ali | **Journal Name**: Australian Endodontic Journal (Impact Factor: 1.5)

2025

An AI-Enabled Framework for Transparency and Interpretability in Cardiovascular Disease Risk Prediction

Authors: Isha Kiran, Shahzad Ali, Sajawal ur Rehman Khan, Musaed Alhussein, Sheraz Aslam, and Khursheed Aurangzeb | **Journal Name**: Computers, Materials & Continua (Impact Factor: 1.7)

2024

Integration of Graph Theory and Brain Microstructural Analysis with Machine Learning for Advanced Diagnosis of Alzheimer’s Disease and Prediction of Cognitive Decline

Shahzad Ali, Wendy Kreshpa, Michele Piana, Matteo Pardini, Sara Garbarino (2024), Integration of Graph Theory and Brain Microstructural Analysis with Machine Learning for Advanced Diagnosis of Alzheimer’s Disease and Prediction of Cognitive Decline, IV Annual Meeting RIN IRCCS Network - Symposia, 21-22 Novermber 2024.

2020

Streamflow forecasting by modeling the rainfall–streamflow relationship using artificial neural networks

Authors: Shahzad Ali and Muhammad Shahbaz | **Journal Name**: Springer's Modeling Earth Systems and Environment (Impact Factor: 2.9)

2019

Entropy-Based Feature Selection Classification Approach for Detecting Phishing Websites

Authors: Shahzad Ali, Muhammad Shahbaz, and Kashif Jamil | **Journal Name**: 2019 13th International Conference on Open Source Systems and Technologies (ICOSST)

2019

Classification of Alzheimer’s Disease using Machine Learning Techniques

Authors: Muhammad Shahbaz, Shahzad Ali, Aziz Guergachi, Aneeta Niazi, and Amina Umer | **Journal Name**: Proceedings of the 8th International Conference on Data Science, Technology and Applications: DATA

2019

Prediction of Diabetes Disease using Data Mining Classification Techniques

Authors: Shahzad Ali, Dawood Saddique, Muhammad Usman Aslam, Muhammad Usman, Umair Maqbool, Shoaib Ejaz | **Journal Name**: Proceeding of Al Yamamah University Engineering Forum (YUENG)

CERTIFICATIONS

06/02/2025

Computational Neuroscience

Link https://www.coursera.org/account/accomplishments/verify/07VQGCCH00F6

01/02/2025

Foundations and Core Concepts of PyTorch

Link https://www.coursera.org/account/accomplishments/verify/G9VTU74YOR5E

23/01/2025

Machine Learning with Python

Link https://www.coursera.org/account/accomplishments/verify/J9W9G33CHJXC

22/01/2025

Data Analysis with Python

Link https://www.coursera.org/account/accomplishments/verify/HPGFVWLFYSGM

21/01/2025

Python for Data Science, AI & Development

Link https://www.coursera.org/account/accomplishments/verify/RBZPAWPJGRXU

16/01/2025

Introduction to Artificial Intelligence (AI)

Link https://www.coursera.org/account/accomplishments/verify/9D0X4HZY1GOZ

16/01/2025

Generative AI: Prompt Engineering Basics

Link https://www.coursera.org/account/accomplishments/verify/ZG0L180E77QP

16/01/2025

Generative AI: Introduction and Applications

Link https://www.coursera.org/account/accomplishments/verify/BUFBQT4LCSAB

05/10/2024

Deep Learning with PyTorch : GradCAM

Link https://www.coursera.org/account/accomplishments/verify/GO97JYGTWI30

30/09/2024

Machine Learning Pipelines with Azure ML Studio

Link https://www.coursera.org/account/accomplishments/verify/QIBD5VKE8BJK

30/09/2024
Build your first Machine Learning Pipeline using Dataiku

Link https://www.coursera.org/account/accomplishments/verify/B6M0FI4FC5K4

30/09/2024
Overview of Data Visualization

Link https://www.coursera.org/account/accomplishments/verify/IU3SPD7UL5TZ

30/09/2024
Create a Profile and Network on LinkedIn

Link https://www.coursera.org/account/accomplishments/verify/RMNAR1WPNR74

30/09/2024
Create Your First Python Program From UST

Link https://www.coursera.org/account/accomplishments/verify/6QVGDSGKCR83

HONOURS AND AWARDS

02/2016
Punjab Educational Endowment Fund (PEEF) Scholarship – Government of Punjab, Pakistan

Talent-based scholarship awarded during Master's degree

Link https://peef.org.pk/

08/2014
Final Year Project (Visible Light-Based Communication System for High Data Rate Real-Time Applications) Funded by ICT R&D Program – Funded by ICT R&D Program

09/2011
ICT R&D Scholarship – ICT R&D, Federal Government of Pakistan

Merit-based scholarship awarded during Bachelor's degree

CONFERENCES & SEMINARS

22/05/2025 – 22/05/2025 Genova, Italy
Update on the Pathophysiology and Neuroimaging of Amyloid-Related Imaging Abnormalities (ARIA)

Attended the seminar “Update on the Pathophysiology and Neuroimaging of ARIA (Amyloid-Related Imaging Abnormalities),” focused on the neuropathology, immunopathological mechanisms, and MRI characteristics of ARIA-E and ARIA-H in Alzheimer’s disease patients undergoing monoclonal antibody immunotherapy.

Link https://registration.symposiacongressi.com/Login.asp?iCurPage=1&IDEvento=286

21/01/2024 – 26/01/2024 Geilo, Norway
2024 Geilo Winter School on the topic of Graphs and Applications

2024 Geilo Winter School on the topic of Graphs and Applications, which took place in Geilo (Norway) from January 21-26, 2024. The event consisted of multiple lectures, a computational competition, and a poster session, totalling 21 hours of instruction and work.

Link https://www.sintef.no/projectweb/geilowinterschool/2024-winter-school/

25/10/2023 – 27/10/2023 Genova, Italy
IEEE Communication Society (ComSoc) eHealth Technical Committee (TC) PhD School

Attended the IEEE Communication Society (ComSoc) eHealth Technical Committee (TC) PhD School held in Genoa, Italy. The school consisted of 15 hours of Lectures with a Final Test.

Link http://130.251.18.53/IEEE_ComSoC_PhD_School/ROOT/pages/index.html

10/07/2023 – 15/07/2023 Košice, Slovakia

Eastern European Machine Learning (EEML) Summer School

Attended the Eastern European Machine Learning (EEML) Summer School held at the Technical University of Kosice, Slovakia, 10-15 July. The school consisted of 40 hours of Lectures and hands-on sessions.

Link https://www.eeml.eu/previous-editions/eeml2023

10/03/2019 – 14/03/2019 Islamabad, Pakistan

Data Science and Machine Learning for Physicists

Invited Speaker at **International School on Physics & Allied Disciplines** at the **National Center of Physics**, Islamabad, Pakistan

TECHNICAL SKILLS

Programming Languages

- Python | C++ [Expert]
- R | Java [Intermediate]

Machine/Deep Learning Frameworks and Tools

- PyTorch [Intermediate]
- Tensorflow | Keras |OpenCV [Intermediate]
- Sklearn [Expert]
- Jupyter Notebook | Google Colab [Expert]
- PowerBI | Tableau [Expert]
- Rapidminer Studio | Weka | KNIME [Expert]

Medical Imaging Tools

Hands-on with neuroimaging tools, specifically for neuroimaging and analysis including, FreeSurfer, MRtrix3, FSL, SPM, Clinica, Nilearn.

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C1	C1	C1	C1

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user

RECOMMENDATIONS

will be furnished upon request.

Genova , 26/09/2025